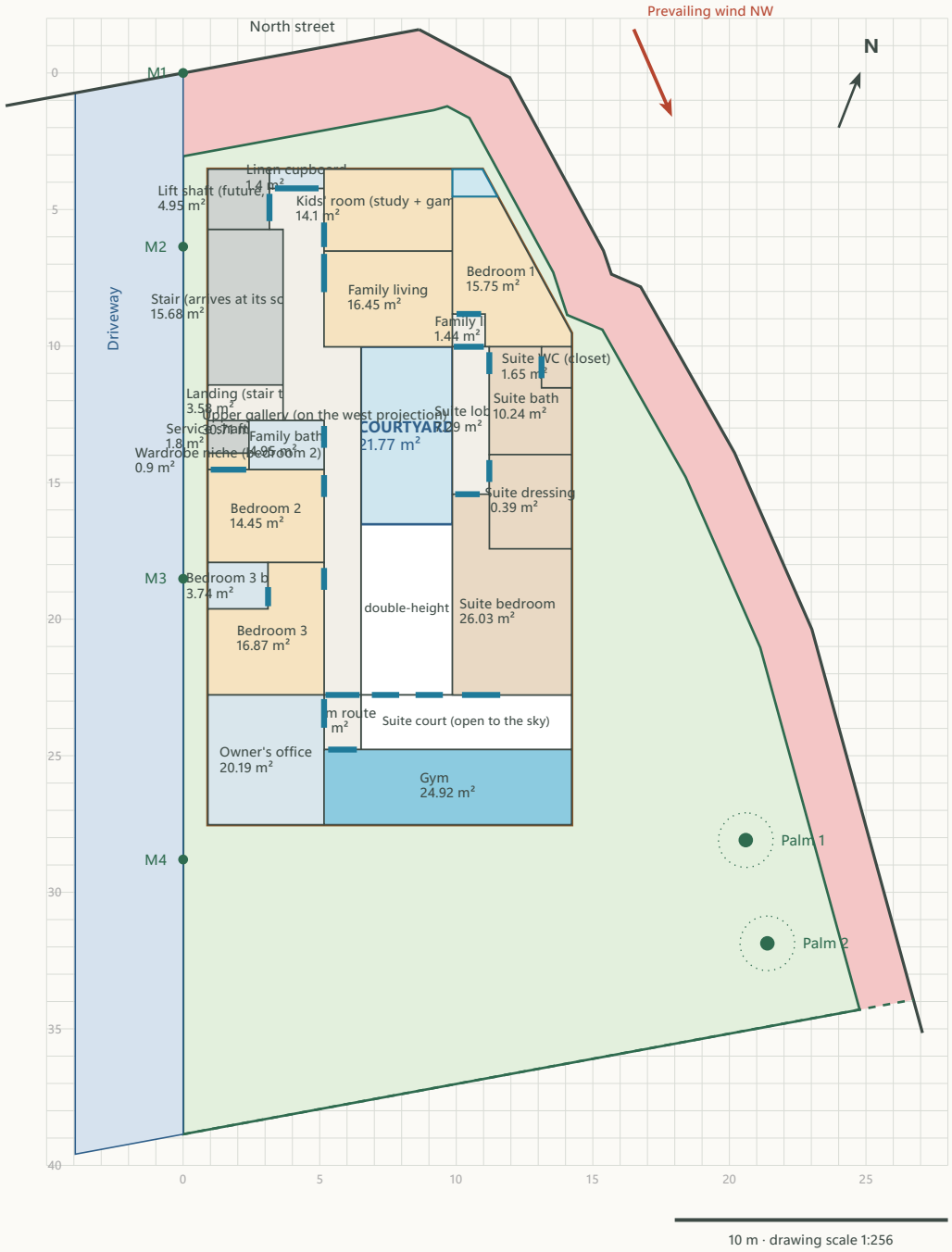


Taif / Palm House

First floor V5.0 · family part + the L (office, gym, suite court, master suite)

20 Oct 2026 · on ground floor V5.0 · rooms in data/rooms_ff.csv · design and furniture in studies/ff_family.typ and studies/ff_suite.typ · schedule and light check on the next pages

First floor V5.0. The gallery runs on a **1.35 m projection** along the west edge of the courtyard and the double-height void (x 4.25–5.6, edge beam on three columns), so the west rooms take the full 4.25 m band: **family bath**, **bedroom 2**, **bedroom 3 with its own bath**. North band: **kids' room (study + gaming)** and the **family living** on the courtyard edge. **Bedroom 1** is one room on the straight y 6.5 line. The **L**: **owner's office** (west), **gym route**, **gym**, the **suite court** open to the sky in front of the clerestory windows, and the **master suite** (lobby, luxury bath with WC closet, walk-in dressing, bedroom). The east duct is removed. Colours: suite (beige) · family (yellow) · baths, office and linen (blue-grey) · gym (blue) · circulation (light) · fixed (grey) · courtyard, void and court (white / blue). Blue marks: doors and the two clerestory windows. Areas: available 243.17 m² · rooms 227.75 m² (circulation 45.72 m²) · suite court 15.42 m². ✓ **data/rooms_ff.csv matches the two studies room for room.**

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**309.68 m²**
Other open voids	**36.36 m²**
Courtyard (open to sky)	**21.77 m²**
Rooms placed (23)	**250.18 m²**
Limit minus courtyard and voids	**251.55 m²**
Air channel + duct	**1.37 m²**
Not yet assigned	**1.37 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Stair (arrives at its south end)	M	2.75 × 5.7	15.68 m²
Lift shaft (future, 2 stops)	M	2.25 × 2.2	4.95 m²
Service shaft (west)	M	1.5 × 1.2	1.8 m²
Landing (stair top)	M	2.75 × 1.3	3.58 m²
Upper gallery (on the west projection)	M	3.35 × 18.55	30.71 m²
Linen cupboard	M	2 × 0.7	1.4 m²
Family bath	M	2.75 × 1.8	4.95 m²
Wardrobe niche (bedroom 2)	M	1.5 × 0.6	0.9 m²
Bedroom 2	M	4.25 × 3.4	14.45 m²
Bedroom 3 bath	M	2.2 × 1.7	3.74 m²
Bedroom 3	M	4.25 × 4.85	16.87 m²
Kids' room (study + gaming)	M	4.7 × 3	14.1 m²
Family living	M	4.7 × 3.5	16.45 m²
Family link	M	1.2 × 1.2	1.44 m²
Bedroom 1	M	4.361 × 5.5	15.75 m²
Gym route	M	1.35 × 2	2.7 m²
Owner's office	M	4.25 × 4.75	20.19 m²
Gym	M	9.061 × 2.75	24.92 m²
Suite lobby	M	1.35 × 5.4	7.29 m²
Suite bath	M	3.011 × 3.95	10.24 m²
Suite WC (closet)	M	1.1 × 1.5	1.65 m²
Suite dressing	M	3.011 × 3.45	10.39 m²
Suite bedroom	M	4.361 × 7.35	26.03 m²

First floor V5.0 · room schedule

wall centre lines (axes) · clear ≈ axes minus half a wall on every edge (from the studies) · CSV coordinates

Room	Part	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	family	2.75 × 1.3	3.57	3.14
Upper gallery (on the west projection)	family	polygon	30.71	27.65
Linen cupboard	family	2 × 0.7	1.4	–
Family bath	family	2.75 × 1.8	4.95	4.04
Wardrobe niche (bedroom 2)	family	1.5 × 0.6	0.9	–
Bedroom 2	family	4.25 × 3.4	14.45	12.84
Bedroom 3 bath	family	2.2 × 1.7	3.74	2.92
Bedroom 3	family	polygon	16.87	14.97
Kids' room (study + gaming)	family	4.7 × 3	14.1	12.44
Family living	family	4.7 × 3.5	16.45	15.16
Family link	family	1.2 × 1.2	1.44	1.08
Bedroom 1	family	polygon	15.75	13.83
Gym route	L	1.35 × 2	2.7	2.03
Owner's office	L	4.25 × 4.75	20.19	18.16
Gym	L	9.06 × 2.75	24.92	22.26
Suite court (open to the sky)	L	7.71 × 2	15.42	–
Suite lobby	L	1.35 × 5.4	7.29	5.94
Suite bath	L	polygon	10.24	8.79
Suite WC (closet)	L	1.1 × 1.5	1.65	1.09
Suite dressing	L	3.01 × 3.45	10.39	9.01
Suite bedroom	L	polygon	26.03	23.56
Rooms (indoor)			**227.75**	
Suite court (open)			**15.42**	
Stair, lift, west shaft, air tower			23.8	

Key numbers

- **Master suite:** 55.6 m² indoors (lobby, bath + WC closet, dressing, bedroom), + the suite court 15.42 m².
- **Circulation:** 45.72 m² (landing, gallery incl. the projection, family link, gym route, suite lobby).
- **Courtyard sky** 3.35 × 6.5 m and **double-height void** 3.35 × 6.25 m on axes (both were 4.70 m wide before the projection); the ground-floor courtyard and living keep their full floor.
- **Bedrooms 2 and 3:** 4.25 m wide on axes (≈ 4.0 m clear).

Where the detail is

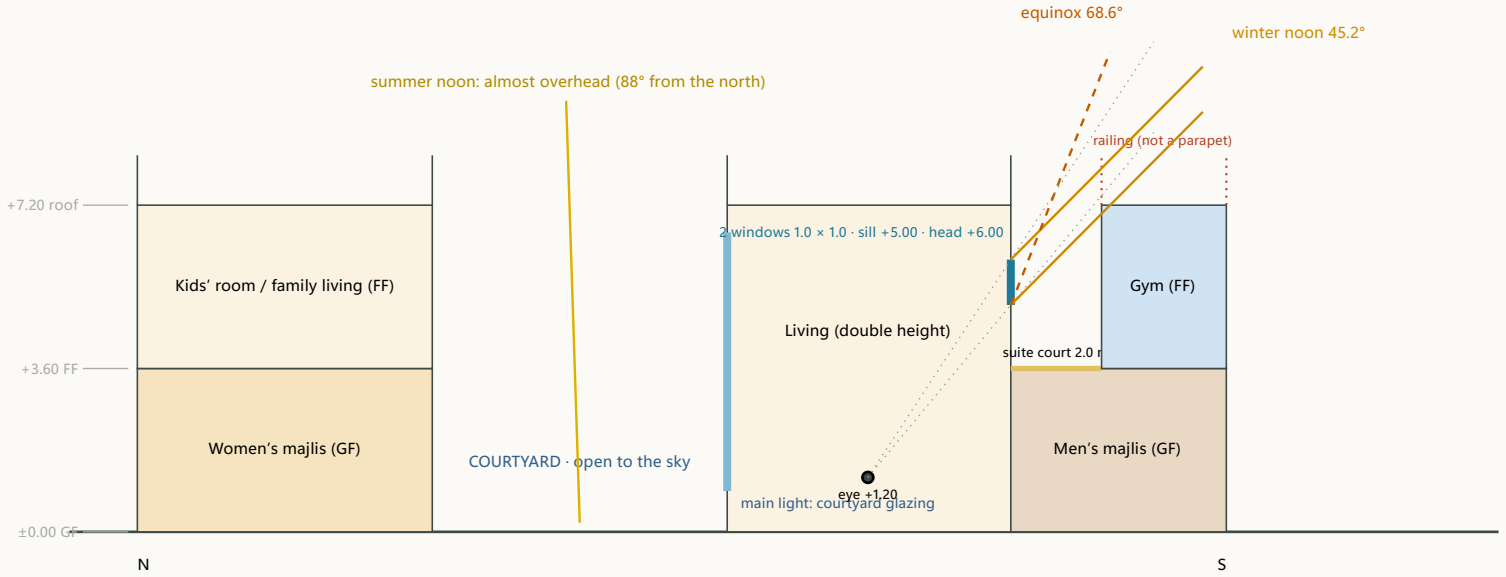
- **Furniture, doors, windows, drains, walking distances, routes ≥ 1.0 m and room checks:** studies/ff_family.typ (part 1) and studies/ff_suite.typ (part 2).
- **Construction notes** (projection, columns, east duct, sunken baths, suite court, cupboards, loads): docs/DETAILS.md.
- **Decisions and open items:** docs/FIRST_FLOOR_BRIEF.md (V5.0) and docs/notes.md.

Open items

- Engineer: the west projection (columns + edge beam, or a plain cantilever), loads and vibration of the gym and walls over the men’s majlis, the y 3.0 partition over the women’s majlis.
- MEP: the stack outside the east wall (suite bath, vent stack), the kitchen extract through the east wall, the hood position south of y ≈ 7.5.
- The clerestory sill vs the suite court floor (eye height, or obscured glass); a daylight check of the narrower courtyard.

First floor V5.0 · light check · the double-height living's south windows

section north → south through the middle band (CSV x ≈ 7.3) · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · parapets 1.10 · a full daylight simulation comes later



Behind the 2.0 m suite court	Winter noon sun on the windows	Equinox noon sun	Sky seen through the windows from the eye
gym with a 1.10 m parapet (top +8.30)	0 %	100 %	42 %
gym with a railing only (solid top +7.20) · **V5.0**	81 %	100 %	100 %

How the percentages are defined. "Sun on the windows": the share of the window height (sill +5.00 to head +6.00) that receives direct sun at solar noon on 21 Dec (winter) or 21 Mar / 23 Sep (equinox), in this section; the ray from each point of the window must pass over the top of the gym's north wall 2.0 m away. "Sky seen": from a seated eye 1.20 m high in the middle of the double-height living floor (CSV y 16.1), the share of the vertical angle of the windows that shows sky rather than the gym. Pure geometry: no reflections, no sky model.

- **The main light of the double-height living comes from the courtyard.** The two small south windows (1.0 × 1.0 m, sill +5.00) add light and let warm air out high on the south.
- **V5.0:** the 2.0 m strip in front of them is now the **suite court**, walled on four sides and open to the sky; the gym's north wall is its south side, ending in a **railing, not a parapet**.
- **Privacy:** the windows open onto the suite court, so their sill stays above eye level from the court floor (or obscured glass); to check against the floor levels.
- **Summer:** the sun is almost overhead; light-coloured court walls and small hoods (≈ 0.4 m) over the windows.